Billing v2 — testing report

Build prompt §13. Updated **01-Nov-2026** (IST). All eight phases: audit, money flow and tax engine (1), documents and checkout (2), ECO compliance — TCS, TDS, GSTR-8, statements (3), books, journals, reports (4), canonical model, sync engine, file connectors (5), GST portal through the GSP (6), live connectors and public API (7), offline counter POS (8).

Local (developer Mac, Postgres.app 18)

Item	Value
Branch	`feat/billing-v2-phase1` (PR #321 into `staging`), from `staging` `2f221d1d`, brought up to `staging` `909de492` (security updates, PR #320)
Commits	phase 1 `d6eac68f`, 2 `8d84ac4c`, 3 `8d1a52ad`, 4 `7641bce2`, 5 `2e558d50`, 6 `e83a51f0`, 7 `bef125a3`, 8 `5489b76a`, scenario 17 test `3c3dfba8`, gap closure (restaurant s.9(5), settlement status, GST type, HSN import) `bf6d71be`, connector guides `1435bdfc`, merge of `staging` `fc9aa17d`, final test fixes (this report's commit)
Migrations	0125–0135 (seller: 0125, 0128, 0130, 0131, 0134; core: 0126, 0127, 0129, 0132, 0133, 0135), each with `scripts/rollback-01xx.sql`
Layouts	single database, and the two-database split (core + seller, built with `scripts/db-split/test-layout.ts`)
Typecheck / lint	`tsc --noEmit`: no new errors (pre-existing errors in `scripts/backfill-delivery-partner-kyc.ts` and `src/server/pmd/sources/xlsx.ts` are on `staging` too); `eslint` clean on every changed file
GSP / IRP	the built-in mock GSP (no GSP contract yet — BLOCKERS B1)

Must-cover scenarios (§13)

#	Scenario	Tests	Single	Split
1	Intra-state B2C (CGST+SGST); inter-state B2B (IGST) with buyer GSTIN	`tax-engine` "whole bill 1a, 1b", "line tax"; `billing-documents` "B2B GSTIN at checkout and the invoice"	Pass	Pass
2	Composition → bill of supply; inter-state blocked	`tax-engine` "2", "order guard"; `billing-documents` "checkout guard"; `billing-gsp` "CMP-08"	Pass	Pass
3	Enrolment-ID → no TCS, intra-state only, still in GSTR-8	`tax-engine` "3", "TCS"; `billing-money-flow` "an Enrolment-ID shop"; `billing-compliance` "GSTR-8 … Enrolment-ID suppliers with no TCS"	Pass	Pass
4	Delivery GST 31 Oct 18% / 1 Nov 5% without ITC in GoKesari's GSTIN; registered rider → none	`tax-engine` "delivery fee s.9(5)"; `billing-money-flow` (2 tests); `billing-documents` "GoKesari's s.9(5) delivery-fee invoice"; `billing-compliance` "Table 15 and 3.1.1(i)"	Pass	Pass
5	Cancellation before packing → full refund, stock restored; partial return → credit note	`cancellation-before-packing` (existing); `billing-documents` "credits the returned lines only". No credit note before packing: no invoice exists yet (DECISIONS D17)	Pass	Pass
6	TCS 0.5% on net after returns; GSTR-8 JSON validates	`tax-engine` "TCS s.52"; `billing-compliance` "TCS follows the money-flow record", "GSTR-8 … validates; filed once"	Pass	Pass
7	TDS 0.1%; ₹5 lakh exemption with PAN; 5% without PAN	`tax-engine` "TDS s.393 scope"; `billing-compliance` "two shops of one individual share the yearly limit"; existing payout TDS tests	Pass	Pass
8	Wallet deductions → GoKesari tax invoice → shop's 2B / IMS flow	`billing-compliance` "the shop's statements"; `billing-gsp` "IMS and the cash ledger"; `billing-books` "posts … the payout's fees and taxes" (fee invoices per payout cycle, D20)	Pass	Pass

#	Scenario	Tests	Single	Split
9	Tally Excel + `.tsf` + XML; round trip; re-export does not duplicate	`billing-sync` "Tally: exports XML + Excel + mapping; the XML reads back the same; re-export reports unchanged"; `billing-live` "Tally connector: two-way books"	Pass	Pass
10	Tally / Busy / Vyapar / myBillBook files → masters, vouchers, stock	`billing-sync` "imports from Busy, Vyapar, myBillBook and Tally Excel" (4 tools)	Pass	Pass
11	Zoho Books and Odoo push and pull	existing `accounting-integration` (mocked APIs); `billing-live` "each connector flag switches on its own provider". Against the vendors' sandboxes: not run (B6)	Pass (mocked)	Pass (mocked)
12	Same voucher edited in GoKesari and Tally → conflict queue	`billing-sync` "conflicts"	Pass	Pass
13	GSTR-1 (Tables 12, 13, 14) validates; 2B pull and IMS; GSTR-3B	`gst-compliance` (existing + ECO rows); `billing-documents` "GSTR-1 counts it as cancelled"; `billing-gsp` "GSTR-1 on the portal", "GSTR-3B … IFF", "IMS"	Pass (mock GSP)	Pass (mock GSP)
14	IRN generate and cancel; e-way bill generate and cancel	`billing-gsp` "e-invoice and e-way bill" (3 tests); existing e-invoice tests	Pass (mock IRP)	Pass (mock IRP)
15	GST session expiry → re-link prompt; GSP down → retry and manual fallback	`billing-gsp` "session expiry and GSP down" (2 tests)	Pass	Pass
16	Offline POS → sync without number gaps or duplicates	`billing-pos` (3 tests: out-of-order and repeated / parallel sync, removed item, price and stock warnings, API)	Pass	Pass
17	Shop A cannot read shop B; dealer cannot see accounts	`billing-security` (another owner, dealer, operator, limited operator, customer, rider refused on 6 billing endpoints; public-API token sees only its shop)	Pass	Pass

Also covered: flags off → nothing changes (every billing test file has an "off" case), immutable issued documents (database triggers), shop documents (credit / debit notes, vouchers, challan, estimate), thermal receipt / JSON / UPI QR / WhatsApp, books posting and reversal, journals / contra / stock journal, marketplace tax report, compliance calendar and reminders, public REST API with OAuth2 scopes and revocation, daily billing cron. Gap closure (`billing-restaurant`, `billing-money-flow`, `billing-compliance`): restaurant food under s.9(5) — GoKesari's invoice for the food, the restaurant's bill of supply at the net price, no TCS, GSTR-1 Table 14(b) and 15, GSTR-3B 3.1.1(i), the 5% withheld from the payout, books balanced; money-flow settlement status from paid payouts; the shop's Enrolment-ID / PAN-only declaration; HSN rates imported from CSV.

Full suite (every test file, both layouts)

Run on `fc9aa17d` (all phases, merged with `staging`) plus the T11 fix, 01-Nov-2026 12.17–13.34 IST, on databases of its own: 265 test files, 3,253 tests per layout.

Layout	Passed	Failed	Skipped
Single database	3,248	5	0
Split (core + seller)	3,240	5	8 (`db-split`: single-layout only, by design)

Failed	Why	Result
`product-import-parser` PUZZLES.xlsm, `product-import` "all fields optional" (both layouts)	Pre-existing (T2): fail the same way on a clean `staging` checkout; outside billing	Not billing
`migrate-on-build` (3 tests), and the after-all hooks of `db-split` and `split-migrate`	The developer Mac's Postgres.app stopped trusting the test's child processes mid-run (server log 12.44 and 12.53 IST: *Postgres.app failed to verify "trust" authentication*, *canceling authentication due to timeout* — its permission dialog went unanswered). Environment, not code	Re-run of the four files that create scratch databases (`migrate-on-build`, `db-split`, `split-migrate`, `db-split-switch`), 13.36–13.39 IST: single **26 / 26 passed**; split **18 passed, 8 skipped**

Every billing test file passes on both layouts. Also: `npm run typecheck` clean; `eslint src tests` 0 errors (7 warnings, none in billing files); `npm run build` passes (CI's environment variables). The blocking CI jobs `test` and `test-split` run the whole suite again on GitHub's Postgres after the push.

Defects

#	Found	Status
T1	`TC-B26-E-045` asserted the 0083 seed of `GST_S95_LOCAL_DELIVERY`; 0126 re-dates it	Fixed: asserts 18% to 31 Oct 2026, 5% from 1 Nov 2026
T2	`product-import-parser` "PUZZLES.xlsm … only a name, only a barcode …" and `product-import` "all fields optional" fail	**Pre-existing**: fail the same way on a clean `staging` checkout; outside billing
T3	A GSTIN verified after 0125 would have stayed UNREGISTERED in the engine	Fixed: `effectiveRegistrationType` falls back to `gst_status`
T4	The new immutable-invoice trigger broke two old tests that rewrite issued invoices	Fixed: those tests switch the trigger off around the rewrite only (D18)
T5	`dealer-step1-flags-geo` hard-coded 8 feature flags	Fixed: uses `FEATURE_FLAG_KEYS`
T6	Production answers only the every-minute / hourly / daily cron routes, so a separate billing cron would never run	Fixed: billing work runs inside the daily job (`billing-daily`, D30)
T7	The test site's basic auth would have blocked the bearer-token APIs	Fixed: `/api/connector/v1/` and `/api/public/v1/` exempt (D29)
T8	Sync fingerprints differed between export and import (one hashed the whole record)	Fixed: shared party / item / voucher fingerprints
T9	The mock IRP refuses 25-character LEGACY invoice numbers	Open as **B4**: e-invoicing needs GST16 numbering on the shop
T10	Two parallel syncs of one offline bill both reported CREATED (one invoice issued)	Fixed before commit: the result says which call issued it
T11	`migrate-on-build` "two migrations started at once" stopped with *out of shared memory*: before the test it dropped the fully migrated schema in one transaction, and with 0125–0135 that needs more locks than Postgres's default lock table (`max_locks_per_transaction` 64; CI uses the same default). Building the schema from scratch takes 6,982 locks on `staging`'s 0124 and 7,159 with 0135	Fixed: the test starts from a new scratch database instead. The migrations themselves are unchanged and apply in one transaction as before
T12	`restaurant` test left `GST_S95_RESTAURANT` switched on for the next test files	Fixed: the test restores the statutory row before and after

Test (test.gokesari.com)

Item	Value
Merge	PR #321 into `staging` 01-Nov-2026 14.36 IST, merge commit `3a0ebcc5`; CI `test`, `test-split` and `build` passed on `9ca2385a` first; no other release in flight
Deploy	Hostinger build of `3a0ebcc5` for test.gokesari.com completed 14.41 IST
Migrations	"Test database" workflow (run 41291657062): split layout, **11 migrations applied (0125–0135)**, 0 repairs, 144 recorded; backup and verify passed, 14.46 IST. Staging gap (code live before its migrations): about 5 minutes, as DEPLOYMENT.md expects
Post-deploy check	`scripts/release/post-deploy-check.sh https://test.gokesari.com`: 5 healthy rounds in a row, 14.46–14.50 IST

Billing test on staging, 01-Nov-2026 14.55–16.53 IST

The five flags billing_v2, tcs_engine, tds_engine, delivery_9_5 and tally_connector were switched on for Pune (MH-PUN) only, 14.56–15.27 IST, for the E2E shop "E2E Four Features Bakery" (PIN 411005, not GST-registered), then restored. Before and after, all 11 billing flags were off for every district (0 differences). Evidence: `test-results/qa/billing-v2-staging-2026-11-01/` (log, screenshots, PDFs, JSON).

Step	Result
Flags off → billing pages and APIs not available	Pass (`pos/items` 404; `/shop/accounts/pos` → `/shop/accounts`)
Shop screens: `/shop/accounts` and its 17 sections, desktop and phone	Pass
Counter bill S2627-00002, 80 mm receipt, UPI QR (sandbox VPA `testsuccess@gocash`), JSON	Pass; findings D-113, D-114, D-115
Estimate ES2627-000001	Pass
POS online (S2627-00003) and offline: saved on the device, synced as S2627-00004, two re-sends → duplicate, no gap	Pass
Prepaid order DB-20261101-4NGI5U (Cashfree sandbox, "Pay (success)"), delivered	Pass: shop's bill of supply BKS-000021/2026-27/000019 (A4, 80 mm, JSON); GoKesari's s.9(5) invoice GKS/2627/000001 (in GSTR-1 Table 15); money-flow row in `/admin/compliance` ; findings D-112, D-116, D-117
COD order	Blocked: the test customer has 2 old open COD orders (limit 2) — staging data (E-16)
Tally export	Fail: refused until 126 items get HSN and GST rate, though the shop is not GST-registered (D-111)
Reports: trial balance, balance sheet, stock, MTR, TCS/TDS credit, P&L	Pass (books balanced at ₹3,852)
GSTR-1 on the mock GSP	Blocked: the E2E shop has no GSTIN (the page says so, 422)
Statements: wallet, settlement, TCS/TDS (PDF and CSV)	Pass
Security: signed out 401; customer, rider-role user, operator 403; owner and admin 200	Pass (rider and limited-operator accounts had no saved session)
District limit: billing_v2 set to Mysuru (KA-MYS) only, Pune shop still opened the POS	**Fail (D-110, High)**
Regression with flags off (guest + signed-in, desktop + phone, 15.28–16.43 IST): 435 cases, pass 258, fail 56, blocked 359, skipped 15 (31-Oct: 264 / 58 / 351 / 15)	No failure caused by billing v2 behaviour; new fails are staging data (E-16, E-17) and two sweeps that listed the bearer-token API (by design, D29) and the DigiLocker mock (E-15)

Defects found, and what was done:

#	Severity	Found	Fix
D-110	High	Billing flags ignored their district list: every check asked "on anywhere", so a pilot switch-on would have started TCS / TDS / s.9(5) for every shop in India	New `isFeatureEnabledForShop` / `requireFeatureForShop` / `enabledFeatureKeysForShop` (the shop's PIN → district): the accounts API, every `/shop/accounts` page, the shop's menus, money flow, s.9(5), payout TCS / TDS (per shop), books posting, statements notices, order checkout, invoices, credit notes, public API, Tally connector, e-invoice and e-way bill routes
D-111	High	Tally export refused a shop outside GST until every item had HSN and GST rate	HSN, GST rate and UQC are asked of a GST-registered shop only; unbalanced vouchers and unknown ledgers still block
D-112	Medium	The customer could not reach GoKesari's s.9(5) invoice	"Delivery fee invoice (GoKesari)" / "Food invoice (GoKesari)" links on My orders and the order page
D-113	Medium	A shop outside GST: counter bill page and A4 PDF said "Tax invoice", the receipt "Bill of supply"; the customer's link said "Tax invoice"	One rule (`counterBillKind`) for the page, A4 PDF, receipt and MTR; the customer's link names the issued document

#	Severity	Found	Fix
D-114	Medium	No screen field for the shop's UPI id	"UPI id for receipts" under Accounts → Settings (billing v2)
D-115	Low	No WhatsApp share; receipt and JSON not linked	"80 mm receipt", "JSON" and "Share on WhatsApp" on the counter bill and the order's invoice page
D-116	Low	A bill of supply said the ECO collects TCS under s.52 and "Prices are inclusive of GST"	Those notes on tax invoices only
D-117	Low (pre-existing, 6 Oct)	"Paid by customer (GoKesari wallet)" for an online payment	"online payment" / "GoKesari wallet" / "cash on delivery"

Not code: E-15 (QA sweeps now list `/api/public/v1/` as token-protected and the DigiLocker mock as a test-site stand-in), E-16 (staging data: COD limit, drained wallet, stale orders, rider and limited-operator sessions), E-17 (SD-01 shop state, SD-05 connection drop).

Fixes on staging, 01-Nov-2026

Item	Value
Merge	PR #323 into `staging` 22.27 IST, merge commit `41ef27cc`; CI `test`, `test-split` and `scope` passed on `9169ed1d` (`test` after one re-run: a rider-withdrawal test that waits for an e-mailed code, in code the PR does not touch, failed once and passed on the re-run and in `test-split`)
Deploy	Hostinger build of `41ef27cc` for test.gokesari.com completed 22.32 IST; no migrations
Post-deploy check	test.gokesari.com: 3 healthy rounds in a row, 22.34–22.35 IST
Retest of D-110 to D-117 on staging	Run 02-Nov-2026 11.17–11.44 IST, after the production release (the owner asked for the release first): all pass, see below

Retest on staging, 02-Nov-2026 11.17–11.44 IST

test.gokesari.com on `41ef27cc`. The same five flags were switched on for Mysuru (KA-MYS) only, then for Pune (MH-PUN) only, then restored: 19 flags compared before and after, 0 differences, all 11 billing flags off. Evidence: `test-results/qa/billing-v2-staging-retest-2026-11-02/` (log, 50 screenshots, 39 files); collected run `docs/testing/qa-runs/2026-11-02-staging-billing-v2-retest.json`.

#	Result
D-110	Pass. Flags on for KA-MYS only: the Pune shop gets 404 on all 7 billing APIs tried; POS, documents, journals, statements and sync pages go to `/shop/accounts`; no billing tiles, menu item, UPI field or bill links on desktop or phone. Flags on for MH-PUN: all served
D-111	Pass. Export check: 0 fixes needed (was 254); the Tally zip downloads from the screen and the API; 6 vouchers, all balanced
D-112	Pass. "Delivery fee invoice (GoKesari)" on the order page and My orders opens GKS/2627/000001 and GKS/2627/000002 (PDF, 200). customer2 404, shop 404, signed out 401, operations 200 (by design)
D-113	Pass. S2627-00001 (cancelled) to -00004: "Bill of supply" on the page (desktop and phone), A4 PDF, 80 mm receipt, MTR rows and the customer's link

#	Result
D-114	Pass. "UPI id for receipts" saved as `testsuccess@gocash` puts a UPI QR on the receipt (`upi://pay?pa=testsuccess%40gocash…`); cleared afterwards
D-115	Pass. "80 mm receipt", "JSON" (200) and "Share on WhatsApp" (wa.me link naming the document) on the counter bill and the order invoice pages; none on the cancelled bill
D-116	Pass. New prepaid order DB-20261102-ERAAO8 (₹85, built-in sandbox "Pay (success)", delivered 11.38 IST): its bill of supply BKS-000021/2026-27/000020 has neither the s.52 clause nor "Prices are inclusive of GST". BKS-…/000019 keeps its old notes (notes are fixed at issue, by design)
D-117	Pass. Both orders: "Paid by customer (online payment) Rs. 85.00"
E-15	Pass. TC-SEC-001, TC-RBAC-002, TC-AUTH-033 (desktop, 7 rows): 211 pages, 397 write endpoints, 11 public writes and 26 cron endpoints refused

New defects (open):

#	Severity	Found
D-118	Medium (not billing-specific)	API error answers carry no `Cache-Control`; Hostinger's CDN (`server: hcdn`) kept a 404 for about 10–20 s and gave it to the next caller, so the customer briefly got 404 on their own invoice after customer2's refused request. 200 answers are not cached; 401 / 403 were not reused. gokesari.com's 404s also come without `Cache-Control` (checked 02-Nov-2026 as a guest)
D-119	Low	A bill of supply still says "Invoice total (inclusive of tax)" with CGST / SGST rows in the PDF, and "Total (tax inclusive)" on the page

Prod (gokesari.com)

Item	Value
Owner's instruction	"push all code to main" (01-Nov-2026 22.3x IST), then "Yes, migrate and merge"
In flight before	none: no other PR into main, last Post-deploy check (b8531a5a) passed, last Hostinger build of gokesari.com completed
Migrations	"Production database" workflow `migrate`, source `staging` (run 41330029033), 01-Nov-2026 22.39–22.41 IST: split layout, **11 migrations applied (0125–0135)**, 0 repairs, 136 recorded, none missing. Restore point 17:09:58 UTC, LSN `0/19BDDA10` (Neon → Restore, PROD_DB.md). Applied before the merge, as DEPLOYMENT.md requires: the old code ran on the new schema overnight (the migrations only add)
Merge	PR #324 (`staging` → `main`) 02-Nov-2026 08.27 IST, merge commit `85a12aa6`; CI `test`, `test-split`, `build` and `scope` passed on `41ef27cc`
Deploy	gokesari.com serves the new build from 08.32 IST (a new route, `/api/invoices/<id>/json`, answers 401 instead of 404)
Post-deploy check	`scripts/release/post-deploy-check.sh https://gokesari.com`: 5 healthy rounds in a row, 08.32–08.36 IST; "Post-deploy check" workflow run 41380443658 passed
Flags	All billing flags off: `POST /api/public/v1/oauth/token` → 404 "This feature is not available." The release changes nothing for shops and customers until a flag is switched on, which waits for the CA's answers (BLOCKERS B2, B3)

Mobile

The Android and iOS apps are a WebView over the website: every billing screen, including the offline POS once open, reaches them with the web deploy. Bluetooth thermal printing, a native scanner and opening the POS with no network at all need a store release (B7).

Sign-off

Environment	Status
Local	Pass for phases 1–8 and the gap closure, both layouts (full run 01-Nov-2026 12.17–13.34 IST; pre-existing and environment failures listed above)
Test	Pass. Staging test 01-Nov-2026 found D-110 to D-117; fixed (PR #323) and all retested as passing on 02-Nov-2026 11.17–11.44 IST. Open: D-118 (Medium), D-119 (Low)
Prod	Released 02-Nov-2026 08.32 IST with every billing flag off; migrations 0125–0135 applied; post-deploy check passed. Switching billing on waits for the CA (B2, B3); D-118 and D-119 should be fixed first